

Rubus spp. (*Rubus armeniacus*, *Rubus ellipticus*, *Rubus fruticosus*, *Rubus nepalensis*, *Rubus niveus*, *Rubus occidentalis*)

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36.1 Introduction

The *Rubus* is a berry-bearing genus belonging to the family Rosaceae and include nearly about 600 to 800 species across the globe (Schulz and Chim, 2019; Sharma et al., 2020). The genus includes different types of berries such as raspberry, blackberry, red raspberry, strawberry, gooseberry, and boysenberry, etc. (Yilmaz-Akyuz et al., 2019). These berries are consumed as raw or in a processed form such as squash, jams, syrups, juices, and jellies. The berries are the source of primary metabolites such as vitamins, dietary fiber, amino acids, and minerals as well as secondary metabolites like phenolics, anthocyanins, tannin, and flavonoids (Schulz and Chim, 2019). This chapter documents the available literature on the six different species of *Rubus*, which include *R. ellipticus* (Fig. 36.1), *R. nepalensis*, *R. fruticosus*, *R. armeniacus*, *R. niveus*, and *R. occidentalis*.

Literature indicates that the different species of the *Rubus* are extensively used in traditional and ethnomedicine in different diseases such as fever, diarrhea, skin ulcer, anemia, burns, and other diseases (Muniyandi et al., 2019). Species under the genus *Rubus* are known by various local names and synonyms Table 36.1.

These species are large perennial shrubs and possess fruits in the form of berries. The species of *Rubus* vary in life form, height, fruit and leaf types, flowering, and fruiting time (Table 36.2). The berries of these species are edible. Parts of the species such as bark, leaf, root, shoot, and whole plant are used in traditional medicine and have the potential to prevent various diseases (Badhani et al., 2015; Schulz and Chim, 2019). The berries are heart-friendly, have good nutritive value and delicious in taste, and contain many antioxidant compounds such as anthocyanin, flavonoids, vitamins, amino acids, and minerals which work as anti-inflammatory compounds and are useful in many diseases (Table 36.2).

All these species are a source of many pharmacological activities such as anti-inflammatory, antimicrobial, antidiabetic, anti-fertility, antioxidant, and anticancer activity which reflects that genus *Rubus* has profound potential fruit for nutraceutical, cosmeceuticals, and pharmaceuticals. The presence of phytochemicals compounds in the fruits provides an avenue for R&D activities in the food industry nutraceuticals and pharmaceuticals drug development.

36.2 Traditional/ethnobotanical uses

The genus *Rubus* has been regularly used in the traditional medicine system, which is helpful in many health problems. It is used to treat wounds, burns, inflammation, gastrointestinal problems, diabetes, helpful in dawn sickness, aid in childbirth, stimulating the uterus, and relieving menstrual cramps (Hummer, 2010; Rocabado et al., 2008). Various other uses of the different parts of the plants includes; root paste of *R. ellipticus* is used in the treatment of bone fracture, fruits (laxative in nature) in constipation, fruits juice in cholera, intake of 10 to 20g early stage fruit paste in the case of gastritis, antacid, diarrhea and dysentery problems are reported (Kumari et al., 2011; Maity et al., 2004). In addition, roots are used to cure paralysis, blood disorders, wound healing, and ulcer problems (Vadivelan et al., 2009; Wang and Lin, 2000). Capsule

TABLE 36.1 Synonyms and common names of six *Rubus* species. (Source– <http://www.theplantlist.org/>).

S. No.	Family	Rosaceae	
	Name of the species	Synonyms	Common name
1.	<i>R. ellipticus</i> Sm.	<i>R. ellipticus</i> var. <i>ellipticus</i> , <i>R. ellipticus</i> subsp. <i>f. asciculatus</i> (Dothie) Focke, <i>R. ellipticus f. obcordatus</i> Franch.	Australia -Yellow raspberry India - Hinsalu, lalanchu, yellow Himalayan raspberry Nepal - Ainselu Phillippines -Buhadoi, bunut, kokobod Vietnam - ngay long
2.	<i>R. nepalensis</i>	<i>R. nepalensis</i> hort.	Himalayan Creeping Bramble Nepal - Nepalese raspberry, Nhyalang (Sherpa)
3.	<i>R. fruticosus</i> L.	<i>R. fruticosus</i> G.N.Jones, <i>R. fruticosus</i> L. ex Direb., <i>R. f. ruticosus</i> Makino and Nemeto, <i>R. fruticosus f. affinis</i> (Weihe and Nees ex Bluff and Fingerh) Wirtg., etc.	Blackberry Denmark : almindelig brombaer Finland : Karhunvattu, oimuvatukka Germany : Echte Brombeere Poland : Jerzyna, jezyna Sweden : bjoernbaer
4.	<i>R. armeniacus</i> Focke	<i>R. hedyocarpus</i> subsp. <i>armeniacus</i> (Focke) Erichsen, <i>R. macrostemon f. armeniacus</i> (Focke) Sprib.	Himalayan Blackberry and AmericanBlackberry Denmark : armensk brombaer Finland : armeniankarhunvatukka Italy : rovo a pelli rossi Poland : jezynakaukaska
5.	<i>R. niveus</i> Thunb.	<i>R. niveus</i> wal ex G. Don, <i>R. niveus</i> subsp. <i>inopertus</i> Focke, <i>R. niveus</i> var. <i>niveus</i>	Mysore raspberry, Ceylon raspberry, Hill raspberry
6.	<i>R. occidentalis</i> L.	<i>R. occidentalis</i> var <i>flavobaccus</i> Blanch. <i>R. occidentalis</i> Subsp. <i>leucoderis</i> (Dougl. Ex Hook.) Focke, <i>R. occidentalis</i> var <i>leucoderis</i> Dougl. Ex Hook. <i>R. occidentalis</i> var <i>occidentalis</i> , <i>R. occidentalis</i> var <i>pallidus</i> L.H. Bailey	Black raspberry Germany : Framboisier de Virginie

**FIGURE 36.1** No Permission Required. Fruit of *Rubus ellipticus*.

prepared from fresh root tips of *R. niveus* consumed (one pill per day) early morning empty stomach with butter and buffalo milk to cure the menstrual cycle (Uniyal et al., 2006). Fruits are also used in dysmenorrhea (Singh et al., 2014), a decoction of the leaves to cure urinary complaints by relaxing uterus muscles, and the fruits in calculi problems (Khajuria and Bisht, 2016; Singh, 2008). Likewise, the *R. fruticosus* plant has many properties such as astringency, mouthwash for spongy gums and ease mouth, swallowing, wounds, diarrhea, respiratory problems, and hemorrhoids (Verma et al., 2014; Zia-Ul-Haq et al., 2014). Verma et al. (2014) reported that the leaf extract used in mouthwash which is useful in gum inflammation, mouth ulcers, sore throat, while, chewing the fresh leaves strengthen the gums and to cure thrush, and fruit juice used to cure asthma. In addition, leaves and roots are a very effective home remedy, which are used to cure anemia, fever, menses, dysentery, and cystitis (Oszmiański et al., 2011). *R. neplansis* fruit are edible and dried powder used in stomach problems,

TABLE 36.2 Morphological and phenological attributes of six Rubus species.							
S. No.	Species name	<i>R. ellipticus</i>	<i>R. nepalensis</i>	<i>R. armeniacus</i>	<i>R. niveus</i>	<i>R. fruticosus</i>	<i>R. occidentalis</i>
1.	Life forms	Perennial	Perennial	Perennial	Perennial	Perennial	Perennial
2.	Growth forms	Evergreen Shrub	Evergreen shrub	Shrub	Shrub	Shrub	Shrub
3.	Height (m)	2–4.5	0.20	3–4.5	1–2.5	3–5	3
4.	Leaf type	Ovate and round shape	Leaves are trifoliate with the three leaflets	The leaves are compound with 3–5, leaflets Sharply toothed	Compound, imparipinnate, 5–11 leaflets.	Compound with three or five oval leaflets	Sessile leaflets
5.	Branch per plant	5–12.5	11.5	na	6.3	na	na
6.	Fruit type	Cluster of droplets	Small and scarlet	Aggregate, shiny, large, black drupelets	Sub globose aggregate of drupelets	Berry	Small and round
7.	Fruit color	Orange – red	Red or oranges	Blackish red	Blackish-brown	Red and black	Black similarly to blackberry
8.	Flowering	April – May	June – July	April – June	March – Sept	March-May	May – June
9.	Flower diameter (cm)	1.99	0.66	2–2.5	0.53	2-3	0.6–0.7
10.	Average number of flower per Truss	11.6	15.2	26.5	10.4	13	7
11.	Fruiting	May – June	July – Aug	Early August	May – Oct	Both season (spring and autumn)	July – August
12.	Fruit yield/branch (kg)	Small-0.095 Medium-0.122 Large-0.175	na	na	na	na	0.0017–0.0025
13.	Fruit length (cm)	0.93	1.55	na	0.79	na	na
14.	Fruit width (cm)	0.85	0.88	na	1.10	na	na
15.	Fruit weight (g)	0.567	1.642	na	0.560	na	na
16.	References	Maikhuri et al., 2009; Sharma et al., 2019 Trivedi et al., 2016	Flora of Nepal, Trivedi et al., 2016	Flora of North America http://www.efloras.org/florataxon.aspx?flora_id=1&taxon_id=242417158	Maikhuri et al., 2009; Quinton et al., 2011 Trivedi et al., 2016	Haq et al., 2014	Dossett, 2011

hypertension, and anemia (Trivedi et al., 2016). Leaf powder of this species is used to cure skin problems, crushed leaf paste heals wounds and works as an antiallergic, the sap of stem and leaf have astringent properties and roots paste used to treat burns and scalds (Bhat et al., 2013; Dutt et al., 2015; Trivedi et al., 2016; Uniyal et al., 2006).

36.3 Distribution and availability

Species of the genus *Rubus* are distributed across the globe, most prominently in the Northern Hemisphere and Indian subcontinent. Briefly, *R. ellipticus* is native to the tropical and subtropical region of the Indian subcontinent, Philippines, and Sri Lanka. The species is often found in along the roadside, dry slopes, mountain valleys, and scattered forests at an elevation of 1000 to 2600 m asl. Likewise, *R. occidentalis* is native to Eastern and Midwestern North America, a locally well-known natural species in Europe, Slovakia, the Czech Republic, Poland, and Russia. The preferred habitat conditions of the species include ravines, edges of woods vegetation having high soil moist percent with well-drained loamy soil (Seeram, 2008; Stoner et al., 2005). *R. fruticosus* is native to Europe with wide distribution reported from Oceania, Asia, North and South America. The species is distributed in tropical and subtropical regions, with an altitudinal between 2000 and 2700 m asl. The species is commonly found along roadsides, hillsides, wasteland, steep banks, scrubby, forest, hillsides, scrub margins, swamps, damp places, creeks, riverbanks, and pasture areas. *R. nepalensis* is endemic to Nepal and found in habitat range in between 2000 and 3000 m. It is widely distributed in Nepal, Western Himalaya, and Eastern Himalaya (Flora of Nepal). Likewise, *R. niveus* is native to India, but has wide distribution reports from Thailand, China, Taiwan, Philippines, Sri Lanka, and Malaysia. The species is commonly found in thickets on slopes, sparse forests, and mountain valleys, at an elevation of 2000 and 3200 m asl, and easily grown in a good well-drained loamy soil. *R. armeniacus* is an American native species, restricted to temperate climates at an altitude of 1800 m asl. Commonly the species is found along roadsides, railway lines, and fence lines. Distributional records of different species are presented (Table 36.3).

36.4 Bioactive/ nutraceutical and nutritional composition

Berries fruits are a rich nutritional, anti-nutritional and mineral source. Among others, the prominent one being crude fiber, sugar, protein, vitamins, nitrogen, calcium, potassium, magnesium, phosphorous, apart from alkaloids, carotenoids, and polyphenols (Schulz and Chim, 2019). Physico-chemical characteristics highlighting nutritional proximate analysis of the fruits of different *Rubus* species are tabulated (Table 36.4). *Rubus* species has been extensively studied for their phytochemical studies. For instance, quantification of phenolic acid such as kamepherol is reported from *R. ellipticus* and *R. niveus* (Muniyandi et al., 2019). Aerial parts of *R. niveus* contain tormentic acid, stigmasterol, quercetin-3-glucuronide, and niga-ichigoside (Tolentino et al., 2015). Badhani et al. (2015) reported the presence of four phenolic compounds (i.e., gallic acid, catechin, chlorogenic acid, and caffeic acid) from fruit extract of *R. ellipticus*. The anthocyanins like cyanidin-3-glucoside, cyanidin-3-rutinoside, pelargonidin-3-rutinoside, peonidin-3-rutinoside, and cyanidin-3-xyloside is reported from *R. armeniacus* and *R. occidentalis* (Lee et al., 2012). Also, both the species *R. armeniacus* and *R. occidentalis* are reported to contain monomeric polyphenols such as ellagic acid, catechin, quercetin, ferulic acid, caffeic acid p-coumaric acid, gallic acid, and flavonol-glycoside (Lee et al., 2012). Leaves and fruits of *R. fruticosus* are reported to contain phenolic acid-like gallic, caffeic acid, and p-coumaric acids, quercetin, hyperoside, kaempferol, myricetin, (+)-catechin, epicatechin gallate, procyanidin B1 and quercetin-3-D-glucoside (Gudej and Tomczyk, 2004; Henning, 1981; Miliwojević et al., 2011). In addition, carotenoids such as α -carotene and β -carotene, zeaxanthin, lycopene, β -cryptoxanthin, lutein, and terpenoids like rubutic acid, rubinic acid, 2- α -Hydroxyursolic acid, and β -amyrin are reported from *R. fruticosus* (Mukherjee et al., 1984; Sarkar and Ganguly, 1978). Anthocyanin like cyanidin-3-O-glucoside, cyanidin-3-O-xyloside, cyanidin-3-O-dioxaloylglucoside, and cyanidin-3-O-(600- malonyl)-glucoside, pelargonidin-3-O-glucoside, malvidin-3-O-glucoside, cyanidin-3-O-arabinoside, cyanidin-3-O-xyloside are reported (Dugo et al., 2001; Fan-Chiang and Wrolstad, 2005; Stintzing et al., 2002; Wu and Prior, 2005). Similarly, the major anthocyanins from *R. occidentalis* include cyaniding aglycone, cyanidin-3-rutinoside, cyanidin-3-xylosylrutinoside, and cyanidin-3-glucoside (Paudel et al., 2014). Besides, anthocyanin like, cyanidin-3,5-di-O-glucoside, cyanidin-3-O-sophoroside, cyanidin-3-O-sambubioside, cyanidin-3-O-glucoside, cyanidin-3-O-rutinoside, pelargonidin-3-O-glucoside, pelargonidin-3-O-rutinoside, cyanidin, and cyanidin-3-O-(2G-xylosylrutinoside) are reported from fruit extract (Krauze-Baranowska et al., 2014). Major bioactive compound present in *Rubus* species is listed in Table 36.5 Fig. 36.2).

TABLE 36.3 Distribution and availability of Six Rubus species.

S. No.	Species name	Country	Region of occurrence	References
1.	<i>R. ellipticus</i>	China	Gansu, Guangxi, Guizhou, Sichuan, Xizang, Yunnan	Flora of China (http://www.efloras.org/florataxon.aspx?flora_id=2&taxon_id=200011503), Flowers of India (http://www.flowersofindia.net/catalog/slides/Mysore%20Raspberry.html), Flora of Nepal (http://www.floraofnepal.org/onlineflora?wildcard_label=Rubus+niveus&wildcard=2810)
		Bhutan	–	
		India	Assam, Sikkim, Meghalaya, Indian Himalayan Region, Kashmir, Kumaon, Nilgiri Hills	
		Laos	–	
		Myanmar	–	
		Nepal	–	
		Pakistan	–	
		Philippines	–	
		Sri Lanka	–	
		Thailand	–	
		Vietnam	–	
2.	<i>R. niveus</i>	Malaysia	–	Flora of China (http://www.efloras.org/florataxon.aspx?flora_id=2&taxon_id=200011503), Flowers of India (http://www.flowersofindia.net/catalog/slides/Mysore%20Raspberry.html), Flora of Nepal (http://www.floraofnepal.org/onlineflora?wildcard_label=Rubus+niveus&wildcard=2810), Plant database (https://plants.sc.egov.usda.gov/core/profile?symbol=RUN14)
		Thailand		
		Sri Lanka		
		Philippines		
		Pakistan		
		Nepal		
		Myanmar		
		India	Assam, Sikkim, Meghalaya, Indian Himalayan Region, Kashmir, Kumaon, Nilgiri Hills	
		Laos		
		Bhutan	–	
		China	Gansu, Guangxi, Guizhou, Sichuan, Xizang, Yunnan	
		Afghanistan		
		Bolivia	Cochabamba, La Paz	
		Ecuador	Loja, Napo, Pichincha, Sucumbíos, Zamora Chinchipe	
		United state		
3.	<i>R. fruticosus</i>	Madagascar		Tropicos (http://legacy.tropicos.org/Name/27805382?tab=distribution)
		South Africa		
4.	<i>R. nepalensis</i>	Nepal	Western Himalaya, Eastern Himalaya	Flora of Nepal (http://www.floraofnepal.org/onlineflora?wildcard_label=Rubus+niveus&wildcard=2810)
5.	<i>R. occidentilis</i>	United State	Arkansas, Colorado, Georgia, Iowa, Kansas, Michigan, Missouri, Nebraska, North Carolina, North Dakota, Oklahoma, South Carolina, South Dakota, Tennessee, Virginia	Tropicos (http://legacy.tropicos.org/Name/27805382?tab=distribution), Plant Database (https://plants.sc.egov.usda.gov/core/profile?symbol=RUN14)
		Canada	New Brunswick, Quebec	
		North America	–	
6.	<i>R. armeniacus</i>	United State		Plants Database (https://plants.sc.egov.usda.gov/core/profile?symbol=RUN14)

TABLE 36.4 Nutritional and phytochemical profile of *Rubus* species.

Parameters	<i>R. ellipticus</i>	<i>R. fruticosus</i>	<i>R. armeniacus</i>	<i>R. nepalensis</i>	<i>R. occidentalis</i>	<i>R. niveus</i>	References
Total phenol	401 mg/g, 35.44 – 80.28 mg GAE/100 g, 1.11 g/kg	20.30 mg/g	–	1.4 g/kg	897 mg/100g	3.21 mg/g, 19.19 – 41.68 mg GAE/100 g, 1.37 g/kg	Saini et al., 2014; Saklani et al., (2012); Badhani et al., 2015; Haq et al., 2014; George et al., 2014; Trivedi et al. (2016); Muniyandi et al., 2019; Belwal et al., 2019; Schulz and chim (2019).
Total flavonoid	215 mg/g, 54.44 – 308.89 mg RE/g	77.77 mg/100 g	–	–	–	4.91 mg/g, 46.67 – 264.76 mg RE/g	
Total anthocyanin	3.18 mg/g, 0.34 g/kg	152 mg/100 g	–	0.9 g/kg	–	5.63 mg/100 g, 0.78 g/kg	
Total tannin	628.32 mg/g, 19.01–66.20 mg GAE/100 g	–	–	–	–	10.50–29.22 mg GAE/100 g	
Carbohydrate	27.12%, 86.4% dry matter	9.61 g/100 g	–	–	–	85.3% dry matter	
Protein	3.68%, 4.4%	–	–	–	–	3.3%	
Ash content	1.3%	–	–	–	–	4.58%	
Vitamin C	1.05%, 0.09 g/kg	21 mg/100 g,	–	0.16 g/kg	–	0.13 g/kg	
Total soluble solids	–	16.2 °Brix	–	–	10.4 °Brix	-	
Total acidity	–	0.5% – 1.4%	–	–	1.0% – 1.6%	–	
Calcium	450 mg/100 g	193 mg/100 g	–	–	–	390 mg/100 g	
Dry matter (%)	33.6	15.0	–	–	–	21.4	
pH	–	3.3 – 3.7	–	–	–	–	
Potassium	6.8 mg/g	8.9 mg/g	–	–	–	7.2 mg/g	

TABLE 36.5 Major bioactive compounds present in *Rubus* species.

Species name	Major compounds	References
<i>R. ellipticus</i>	Gallic acid, kamepherol, catechin, chlorogenic acid, <i>m</i> -coumaric acid, 4-hydroxybenzoic acid, and phloridzin	Muniyandi et al., 2019; Badhani et al., 2015; Belwal et al. (2019), Kewlani et al. (2022)
<i>R. fruticosus</i>	Campesterol, stigmasterol, rubinic acid, rubitic acid , tocopherol, morin, kaempferol, matairesinol, 2,3-O-isopropylidenyl-2 α ,3 α , 19 α -trihydroxyurs-12-en-28- oic acid	Haq et al., 2014; Ono et al., 2016
<i>R. niveus</i>	Kamepherol, tormentic acid , sigmasterol, quercitinglucoronide, and niga-ichigoside	Muniyandi et al., 2019; Tolentino et al. (2015)
<i>R. occidentalis</i>	Ellagic acid, catechin, quercetin, ferulic acid , caffeic acid p-coumaric acid, gallic acid, cyanidine aglycone (Cy), cyanidin-3-rutinoside (Cy 3-rut), cyanidin-3- xylosylrutinoside (Cy 3-xylure) and cyanidin- 3- glucoside (Cy 3-glc)	Lee, Dossett, and Finn, 2012; Paudel et al., 2014
<i>R. armeniacus</i>	Cynidin -3-glucoside, cynidin-3-rutenoside, pelargonidin-3-rutinoside, peonidin-3- rutenoside	Lee, Dossett, & Finn, 2012
No Permission Required.		

36.5 Effect of ripening stages on nutritional attributes

Nutritional contents in the fruits are reported to vary with their ripening stages (Belwal et al., 2019). For instance, changes in polyphenols, anthocyanin and morphological attributes with changing ripening stages of *R. ellipticus* fruit, however, no major change in the fruit length and fruit diameter was recorded (Belwal et al., 2019). Phyto-chemical activity such as total phenolics, total flavonoids, total tannins and antioxidant potential (ABTS) were found higher in unripe fruit of *R. ellipticus*. Similar results are reported for *R. occidentalis*, where the highest antioxidant activity (ORAC) and phenolics value were reported from unripe fruiting stage (Vadivelan et al., 2009; Wang and Lin, 2000).

36.6 Pharmacological activities

36.6.1 Antioxidant activity

Rubus species are enriched with good amount of bioactive phyto-constituents, especially anthocyanins and phenolic compounds, and all these components play an important role in combating various diseases and scavenging ROS species (Trivedi et al., 2016). The fruits extract of ten different genotypes of *R. ellipticus* in acidic methanol medium showed strong antioxidant activity (Badhani et al., 2015). Antioxidant activity using different *in vitro* assays showed varied responses. For instance, ABTS varies from 3.34 to 4.58 mM AAE/100g fw, DPPH from 26.36/100g fw to 27.72 mM AAE/100g fw, and FRAP activity from 1.9 to 3.43 mM AAE /100g fw. Similarly, in the case of *R. niveus* similar responses are reported (i.e., ABTS 2.97 mM AAE/100g fw; DPPH 27.84mM AAE/100g fw and FRAP 2.06 mM AAE/100g fw). Muniyandi et al. (2019) reported comparatively better inhibitory potential (IC₅₀) potential for *R. ellipticus* than *R. niveus*. Besides, the antioxidant potential of *R. occidentalis* and *R. fruticosus* are also reported (Kostecka-Gugała et al., 2015; Schulz and Chim, 2019). Trivedi et al., 2016, reports higher total antioxidant capacity of *R. nepalensis* (4.25 μ mol TE/g fw) as compared to the *R. ellipticus* (2.7 μ mol TE/g fw) and *R. niveus* (2.25 μ mol TE/g fw) while FRAP activity was highest in *R. nepalensis* (35.965 μ mol/g) followed by *R. niveus* (30.25 μ mol/g) and *R. ellipticus* (14.365 μ mol/g).

36.6.2 Antiproliferative activity

ROS species accumulation causes major damage to cells that leads to the prognosis of several degenerative diseases. *R. ellipticus* extract are reported to possess significant scavenging activity. Acetone extract of the fruits of *R. ellipticus* showed significant antiproliferative activity (60%) against the human cervical cancer cell lines (Saini et al., 2014). Similarly, *R. fruticosus* showed high antiproliferative activity using methanol fruit extract (100 μ g/mL) against human cancer cell lines that is, breast (MCF-7), colon (COLO-205), lung (A549), and pancreatic (NCL-H322). Of the different cell lines, lung cell

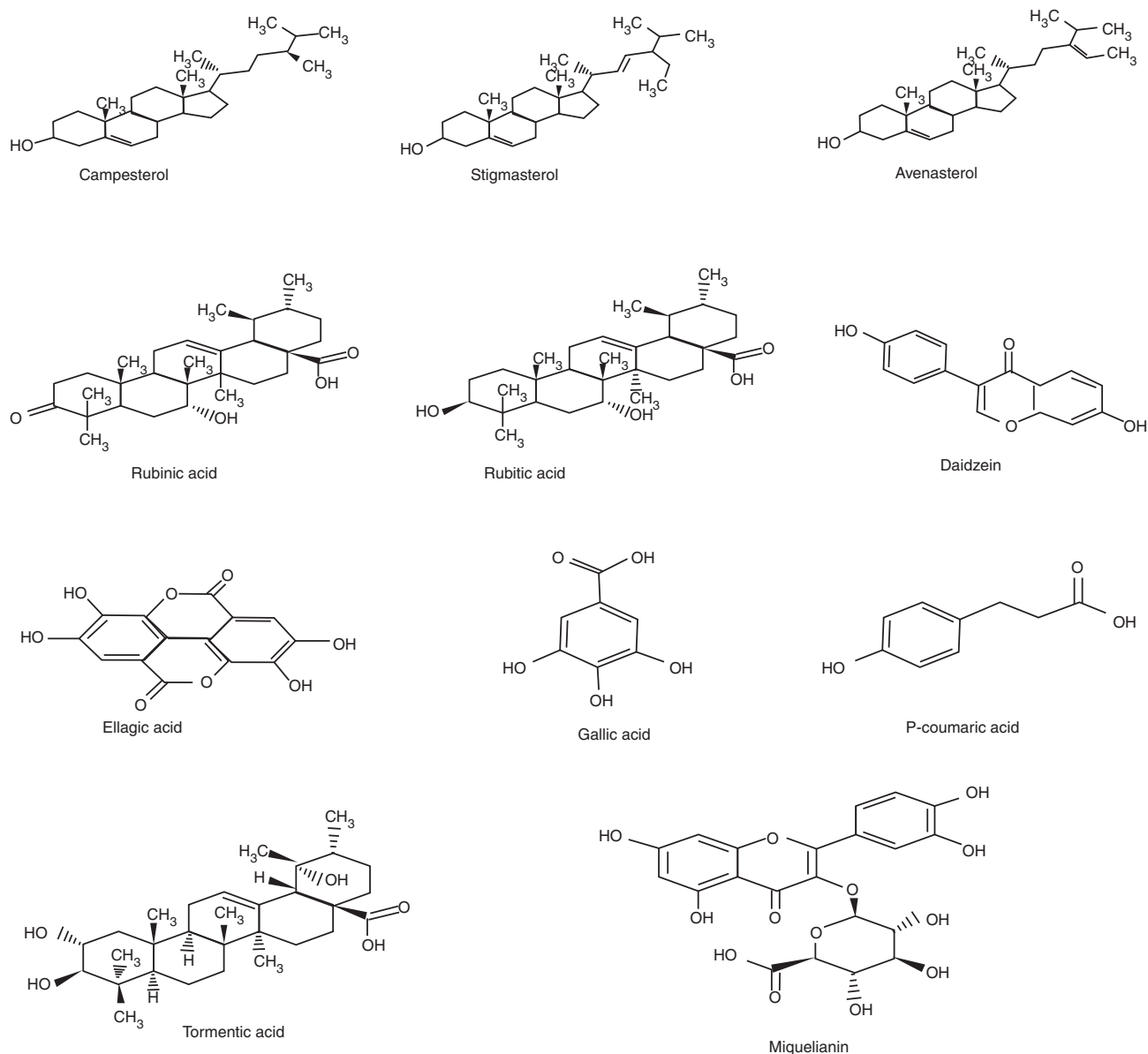


FIGURE 36.2 No Permission Required. Structure of major bio-active compounds present in *Rubus* species.

line was most sensitive, while pancreatic cell line was least sensitive (Bhagat and Thusoo, 2015). Furthermore, fruits extract of *R. ellipticus* and *R. niveus* at a concentration of 10 µg/mL inhibited 50% of cancer cells proliferation in human colon cancer line (Caco-2) (Muniyandi et al., 2019). In case of *R. occidentalis*, extract with concentration of 1.2 mg/mL inhibits HT-29 Colon cancer cell growth from 71% to 118% than low dose concentration 0.6 mg /mL with inhibition ranging between 33 to 82% (Johnson et al., 2011).

36.6.3 Antitumor and wound healing properties

Wounds are associated with the onset of biochemical events such as inflammatory, proliferative and remodeling phases, generating numerous ROS species (MacKay and Miller, 2003). Bioactive compounds from plant extract significantly quench the generated ROS species and thus maintain cell homeostasis. Acetone root extract (250 mg/kg) of *R. niveus* significantly increased antioxidant enzymatic (i.e., SOD, CAT, GPx, and GR) activity on Male Swiss albino mice liver tissue and prolong the life span with Ehrlich ascites carcinoma (EAC) (70.04%) by reducing the volume of Dalton's lymphoma ascites (DLA)

solid tumors. Besides, 2% ointment extract decreased the wound contraction area (91.2%), increased healing tissue tensile strength ($50.63 \pm 1.02 \text{ N/cm}^2$), and reduced *S. aureus* wound infection (80.63%) on 12th day of administration; thus highlighting the potential of extract in wound healing (George et al., 2014). Likewise, methanolic leave extract (250mg/kg) increased the life span of mice with EAC (46.76%) and to reduce the volume of DLA solid tumors (2.56 cm³). The 2% ointment (extract) decreased the wound contraction area by 94.23%, increased healing tissue tensile strength ($42.45 \pm 1.34 \text{ N/cm}^2$), and reduced *S. aureus* wound infection (79.25%) (George et al., 2013).

36.6.4 Antimicrobial/antibacterial activity

The fruits of *R. ellipticus* and *R. niveus* in ethanol extract showed promising antibacterial activity against four microbial strains *Staphylococcus aureus*, *S. epidermidis*, *Pseudomonas aeruginosa* and *E. coli* and four fungal strains *Aspergillus flavus*, *Candida albicans*, *C. krusei* and *Trichoderma lignorum* (Shibu Prasanth and Pratap Chandran, 2017). Likewise, Krauze-Baranowska et al. (2014) investigated fruit extract of *R. occidentalis* against 15 strains of bacteria (both gram positive and gram negative) and two human pathogens *Corynebacterium diphtheriae* and *Moraxella catarrhalis*. The antimicrobial activity of the extract varied according to different microbial strains, however, human pathogens *C. diphtheriae* and *M. catarrhalis* were most sensitive to *R. occidentalis* extracts. fruit, root, leaves, and stem methanolic extract of *R. fruticosus* showed potent antimicrobial potential against *E. coli*, *S. aureus*, *P. aeruginosa* *Salmonella typhi*, *Proteus mirabilis*, *Micrococcus luteus*, *Citrobacter*, and *Bacillus subtilis*. The highest inhibition concentration was recorded in fruit followed by leaves, root and stem, respectively (Riaz et al., 2011).

36.6.5 Anti-inflammatory activity

Rubus species are a rich source of phenolic compounds exerting high antioxidants activity against ROS species. Ellagitannins from *R. fruticosus*, tested under *in vitro* and *in vivo* model are reported to have anti-inflammatory and antihelicobacter pylori properties. Under *in vitro* condition, ellagitannins is reported to exert anti-inflammatory inhibitory effect on cytokines induced transcription of NF- κ B (IC₅₀ 0.67 mg/mL) and IL-8 secretion (IC₅₀ 0.7 mg/mL). Also, ellagitannins inhibits ethanol induced gastric lesion in male Wistar rats and inhibits NF- κ B nuclear translocation and secretion of CINC-1 (IL-8 homologue) (Sangiovanni et al., 2013). Anthocyanins fraction (Cyanidin-3-O-glucoside) is reported to inhibit lipopolysaccharide (LPS) induced NF- κ B activation, I κ B α degradation, decreased ERK-1/2 phosphorylation and NO production (Pergola et al., 2006). Methanolic leave extract (400 mg/kg) of *R. ellipticus* significantly reduced carrageenan and croton oil induced paw and ear edema anti-inflammatory activity in Wistar albino rats (66.47 %) and ear edema in Swiss albino mice (45.78 %) (George et al., 2013). Likewise, George et al. (2014) reported, root acetone extract of *R. niveus* significantly reduced paw edema by 74.52% and ear edema by 54.53% in Wistar albino rats and Swiss albino mice. Similarly, unripe and half-ripened ethanol fruits extracts of *R. occidentalis* reduced NO production, prostaglandin E2 and pro-inflammatory cytokine in LPS stimulated RAW264.7 murine macrophages (Yang et al., 2007).

36.6.6 Antidiabetic activity

The disease is characterized by hyperglycemia, dyslipidemia and protein metabolism resulting in insulin malfunction and increases the risk of cardiac related diseases. The fruit of *Rubus* are reported to have antidiabetic potential (Schulz and Chim, 2019; Sharma et al., 2020; Salehi et al., 2013). For instance, ethanolic extract *R. ellipticus* significantly reduced blood glucose level to $73.5 \pm 2 \text{ mg/dL}$ as compared to normal control ($75.5 \pm 2.4 \text{ mg/dL}$) and diabetic control ($298.0 \pm 3.5 \text{ mg/dL}$) (Sharma and Kumar, 2011). Salehi et al. (2013), reported that methanolic extract of *R. fruticosus* exhibited α -glucosidase inhibitory activity with IC₅₀ value of $0.5 \pm 0.2 \mu\text{g/mL}$ as compared to acarbose ($10.4 \pm 0.7 \mu\text{g/mL}$). Such type of investigation is however, required in other *Rubus* species to harness the full potential in diabetic patient.

36.6.7 Neuro-pharmacological activities

Uses of synthetic psychoactive drug are reported to have illicit addiction with dire consequences such as autonomic, allergic, hematopoietic, and central nervous system (CNS) related disorders. Considering this, methanolic extract of different parts (root, leaves and stems) of *R. fruticosus* was reported to have significant neuro-pharmacological activity on Albino mice model. All extracts administered were found to have anxiolytic and antidepressant effect, while no effect was reported on muscle releasing or sedative effect. Of the different extract prepared highest neuro-pharmacological activity was reported in fruit followed by root, leaves and stem (Riaz et al., 2011).

TABLE 36.6 List of commercially available products containing *Rubus* as an active ingredient.

S. No.	Product name	Price in US\$	Uses	Manufacturing company
1	Organic Blackberry Capsule 60/760 mg	\$38		Arifoğlu , Germany
2	Blackberry leaf Fruticosus	\$29.95	Supports overall colon health	Florida Herbal Pharmacy
3	Blackberry leaf (<i>Rubus fruticosus</i>)	\$9.95		Florida Herbal Pharmacy
4	Red raspberry leaf	\$6.95		Florida Herbal Pharmacy
5	Red raspberry seed oil	\$25.00	Healthy skin and facial moisturizer	Berry beautiful
6	Raspberry red	\$0.6	Food color preparation	Makman, India
7	Black raspberry Jam	\$80.56	Wonderful on Toast, Bagels and Crumpets	Khuffman's fruit farm market, Ohio
8	Wild Blackberry, & Apple Jam	\$9.98		Bumblee's Preserves
9	Black Raspberry Flavored Vodka	\$50.8		Ciroc
10	Raspberry Fruit Tea	\$2.1		OMG Chai, India
11	Raspberry (800 mg) Capsule	\$5.3	Promote weight loss	Himalayan Organics, India
12	Organic Raspberry leaf	\$33.2	Naturally caffeine Free herbal Tea	Traditional Medicinals, India
13	Black raspberry Vanilla Cream	\$33.5		Bath and Body Works
14	Radical raspberry	\$4.7	Source of Vitamin C and anti-oxidant	Flyberry Gourmet
15	Blackberry (<i>Rubus fruticosus</i>)	\$19.95	Alcohol -Free	Hawaii Pharm
16	Blackberry leaf	\$15.99		Herba Organica, England
17	Black berry seed oil	\$19.90	Moisturizes and reduces wrinkles	Botanical beauty, USA
18	Shir- Organic pure Blackberry moisturizer	\$56.00	Healthy skin and fresh	Shira, Egypt
19	Hair food blackberry & Anjou Pear Strength Shampoo	\$38.09		Hair food
20	Fresh Freeze-Dried Black Raspberry	\$12.97		Electic, US

36.6.8 Anti fertility activity

Plant extracts has been long used to cure the plethora of fertility problems in human beings. Reports on ethanolic extract of *R. ellipticus* roots in anti-fertility activity were reported in albino rats (Sharma et al., 1983). Ethanolic extracts produced highest anti-implantation activity in 1 to 7 days of pregnancy. Extracts from aerial parts are reported to completely inhibit likely pregnancy chances in rats. Similar reports are available elsewhere (Dhanabal et al., 2000). Ethanolic leave extract from *R. ellipticus* and *R. niveus* in female Wister albino rat model are reported to elicit 91.43% (*R. ellipticus*) and 84.88% (*R. niveus*) antifertility activity (Dhanabal et al., 2000).

36.6.9 Toxicological activity

The aqueous leaf extract and silver nano particals (AgNPs) prepared from the *R. ellipticus* leaf extract showed higher toxicity (LC₅₀) against the mosquito larvae (AlQahtani et al., 2017). Also, AgNPs elicit zero hatchability at 60, 75, and 90 µg/mL on *A. stephensi*, *A. aegypti*, and *C. quinquefasciatus*, respectively. *R. ellipticus*-fabricated AgNPs were least harmful to other non-targeted aquatic species (AlQahtani et al., 2017). Toxicity of methanolic *R. fruticosus* fruit extract at a concentration of

20 mg/mL was reported to exhibit an excellent anthelmintic activity against *Raillientina spiralis* and *Ascaridia galli* that was 1.37 times higher than albendazole (synthetic analogue) under *in vitro* Brine shrimp cytotoxicity assay. The acute toxicity of the extract was found to be safe up to a concentration of 100 mg/kg (Ali et al., 2013). Similarly, *R. niveus* fruit extract were observed to be safe in acute oral and dermal toxicity up to a concentration of 2g/kg in male Wistar rats (George et al., 2014).

36.7 From field to Industry

Rubus species, including raspberries, blackberries, and dewberries, has drawn wide attention not only being rich in nutritional profile, but also for their health promoting benefits. Besides rich in vitamins, major and minor minerals, sugars, dietary fibers, they are an important source of phenolic acids, flavonoids, anthocyanins, tannins, and ellagic acid conjugates (elagitannins, sanguin H-6, and lambertianin D) (Krauze-Baranowska et al., 2014; Machado et al., 2015; Schulz and Chim, 2019). Usually the fruits are preferred to be consumed fresh, but many of the processed products namely dietary supplements, juices, yoghurts, ice creams, chocolates, jam, jellies and candies, syrups, wines, and beer are available in the markets across the world (Table 36.6). Berries are an excellent source of anthocyanins, and an excellent source of natural dyes for various food supplements, which could be potential replacers of harmful synthetic colorants (Ivanovic et al., 2014). Besides, the processed extracts are versatile and are in use for the preparation of cookies, bread, cakes, pastries, pudding, soups, and salads, etc. (Lim, 2012). Cosmetic ingredients containing root and berry extract are used in phyto-cosmetics product preparations like sunscreens, moisturiser, anti-bacterial and anti-aging creams (www.nrcresearchpress.com). Similarly, the leave extract of *R. fruticosus* are used in the formulation against fever, eyes problem, and obesity and piles problems (www.nrcresearchpress.com; Ivanovic et al., 2014).

36.8 Challenges and future recommendation

Rubus species as a whole are enriched with tremendous nutritional and bioactive compounds. The health benefits potential of these compounds are widely accepted and recognized against various life threatening disorders like, cancer, aging, inflammation, diabetes, cardiovascular diseases, and nervous disorders, etc. Considering the health benefits; products like food supplements, juices, yoghurts, chocolates, jellies and candies, syrups, etc., along with phyto-cosmetic products are commercially available. Fruits of the species are perishable, therefore, needs special care during postharvesting process. Ripening stage is reported to affect the accumulation of various bioactive molecules, emphasizing the importance of harvesting time for extraction of compounds of interest will be useful. Likewise, extraction and purification of bioactive compounds are a critical area of research for product formulation and encapsulation. In depth research in this direction would be helpful to harness maximum benefit by optimizing an extraction process. Furthermore, the studies reported for different diseases need to be tested for clinical trial for validating the effectiveness of *in vitro* and *in vivo* results and its ethnotherapeutic importance.

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Non-Print Items

Abstract

The berries are rich source of nutraceutical compounds and daily intake promotes human health, prevents chronic illness and metabolic disorders. The genus *Rubus* includes 600-800 species, which are mainly distributed in hilly regions of Asia, Europe, Africa and North-America. The berry of the *Rubus* attract attention due to its taste, aroma and health promoting bioactive compounds such as vitamins, fatty acids, amino acids, minerals and polyphenols. The objective of this chapter was to describe a brief ethno-medicinal uses, distribution, bioactive compounds and pharmacology activities of six *Rubus* species such as *R. ellipticus*, *R. nepalensis*, *R. fruticosus*, *R. niveus*, *R. armeniacus*, and *R. occidentalis*. These species possesses various health promoting and disease curing properties such as antioxidant, antiproliferative, antitumor, antimicrobial, anti-inflammatory, anti-diabetic, and anti fertility, etc. The available literature on different *Rubus* species suggests the potential of those species in functional food, nutraceuticals, pharmaceuticals and source of livelihoods.

Keywords

Berries; Distribution; Bioactive compounds; Ethnobotany; Pharmacology